

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 25

GCD

AIM: To write and execute an 8085 assembly language program to find GCD of two numbers in an array and store result in memory.

ALGORITHM

1. Store the first number at memory location **2050H**.
2. Store the second number at memory location **2051H**.
3. Load the first number into register **B** and the second number into register **C**.
4. Compare the two numbers.
5. Subtract the smaller number from the larger number.
6. Repeat the subtraction until both numbers become equal.
7. The equal value is the **GCD**.
8. Store the GCD at memory location **2052H**.
9. Stop the program.

CODE:

```
LXI H,6000
MOV A,M
INX H
MOV B,M
LOOP: CMP B
JZ STORE
JC EXG
SUB B
JMP LOOP
EXG: MOV C.,B
MOV B,A
MOV A,C
JMP LOOP
```

STORE: STA 6009

HLT

OUTPUT:

Registers

A00

BC0000

DE0000

HL1771

PSW0000

PC421B

SPFFFF

Int-Reg00

S0

Z1

AC0

P1

C0

Flag

S0

Z1

AC0

P1

C0

Decimal - Hex Conversion

Decimal

Hex

00

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

2051

-

+

8

Update Memory

Load me at

1LXI H, 6000

2MOV A, M

3INX H

4MOV B, M

5LOOP: CMP B

6JZ STORE

7JC EXG

8SUB B

9JMP LOOP

10EXG: MOV C, B

11MOV B, A

12MOV A, C

13JMP LOOP

14STORE: STA 6009

15HLT

Data

Stack

Keypad

Memory

I/O Ports

Start

2050

OK

Address (Hex)

Address

Data

0802205012

080320518

080420524

080520530

080620540

080720550

080820560

080920570

080A20580

080B20590

080C20600

080D20610

080E20620

080F20630

Line No

Assembler Message

0Program assembled successfully

Simulator: Idle

34°

Search

ENG IN

11:44

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